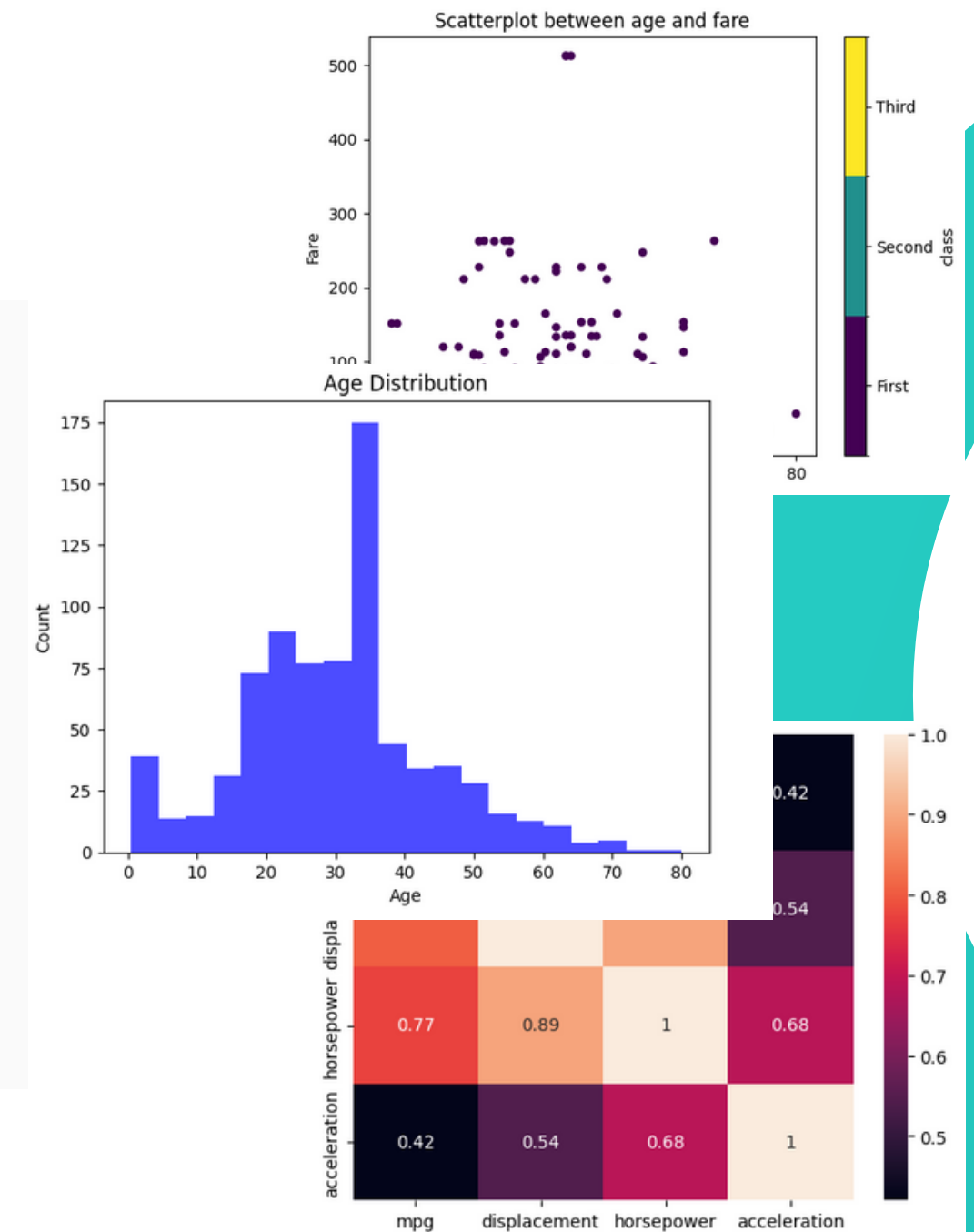
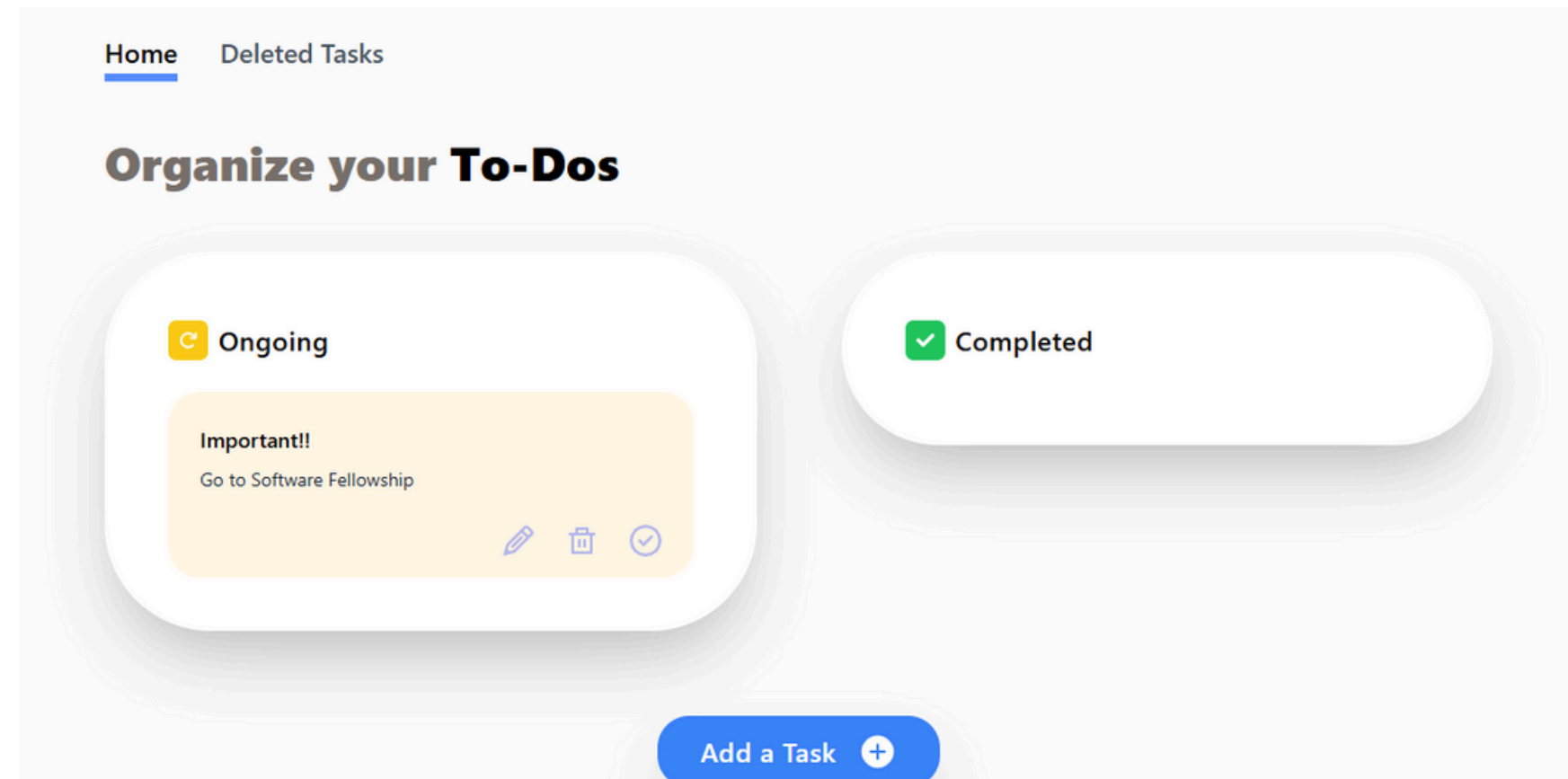
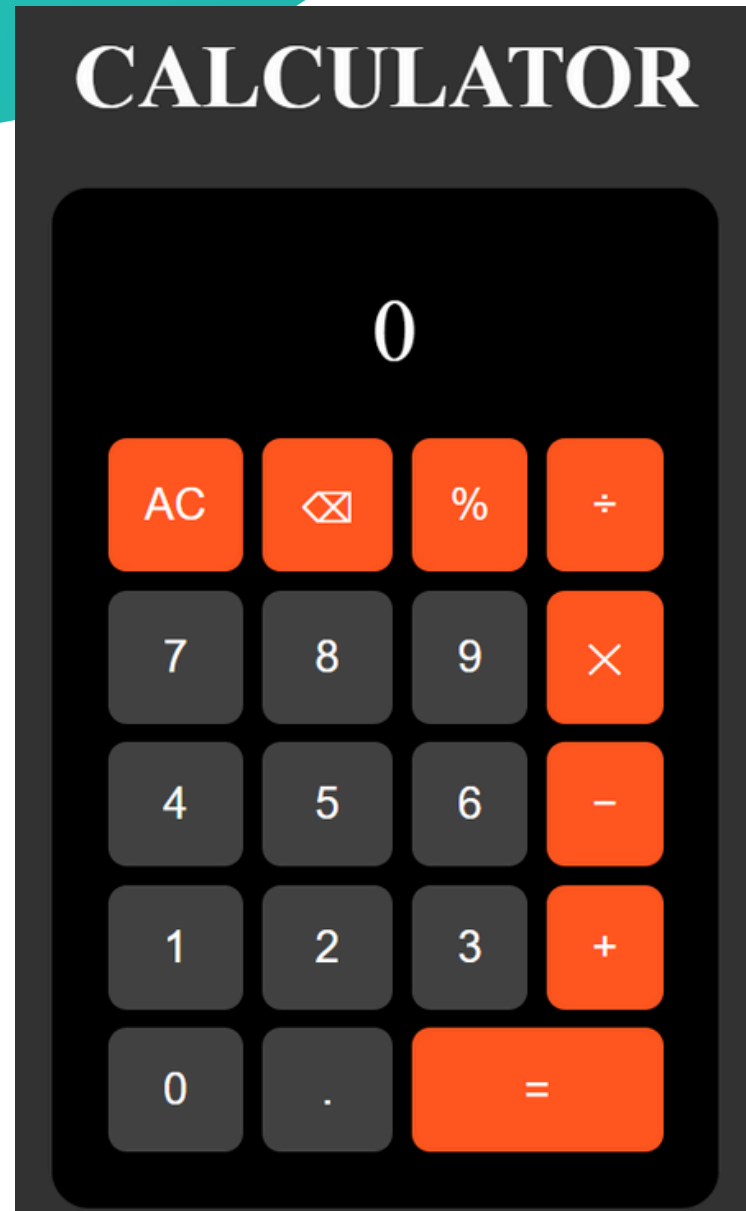


Welcome to Software Fellowship 2025

Learning By Doing



DAY 1



LOCUS

HTML & CSS

Software Fellowship 2025



HTML

Overview

01

About HTML

02

Why Learn HTML

03

HTML Tags

04

Hello world in HTML

05

Assignment

- **Hyper Text Markup Language**
- **Describes content and format of webpages using tags**
- **Standard language for creating web pages and web applications.**
- **Developed by Tim Berners-Lee in 1991.**
- **Foundation of web development.**

HTML



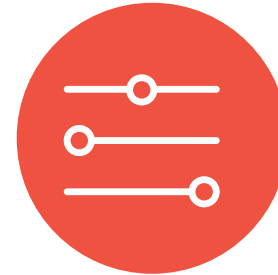


Why Learn HTML?



Foundation of Web Development

HTML is the backbone of all web pages.



Customization and Control

Knowing HTML gives you the ability to customize web content beyond what is possible.



SEO Benefits

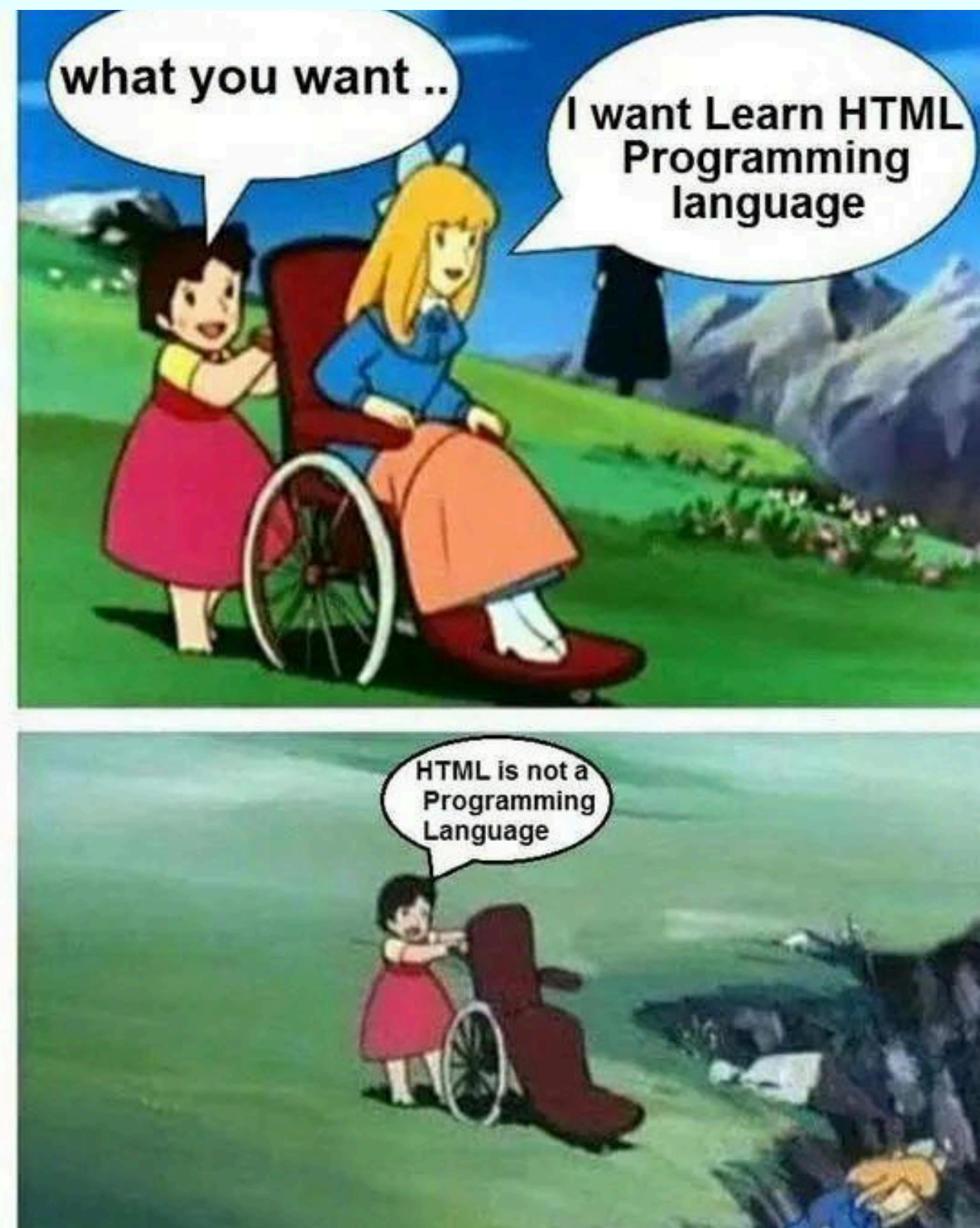
HTML knowledge helps in optimizing web pages for search engines (SEO).



Understanding Web Technologies

HTML, CSS, and JavaScript together create modern web experiences. Learning HTML is the first step to understanding this trio.

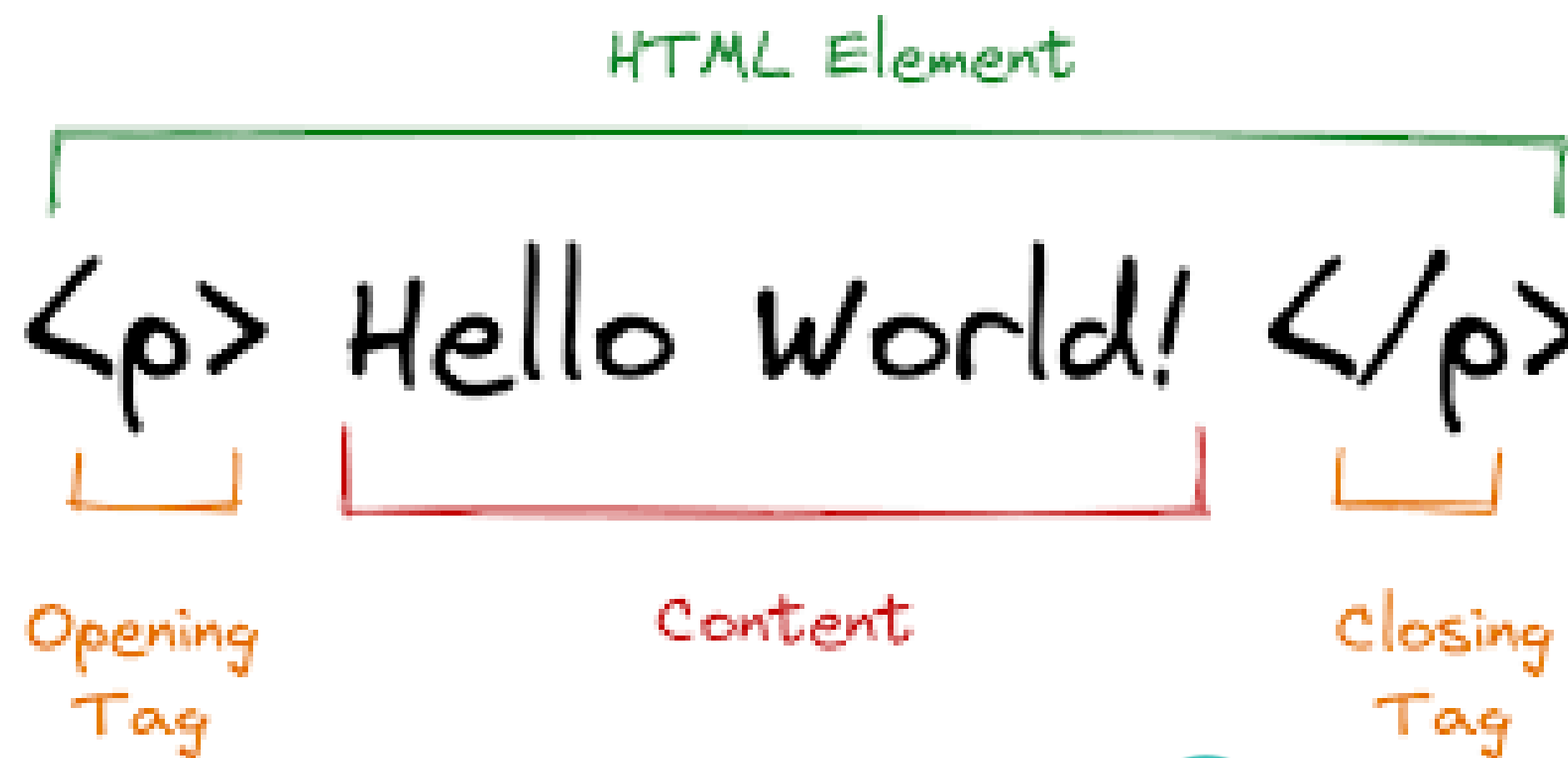
HTML is not a Programming Language !!



HTML Tags

- Define how your web browser must format and display your web page
- Most tags come in pairs: an opening tag `<tag>` and a closing tag `</tag>`.

Example:



Common HTML Tags

- **<html>**: Root element of an HTML document.
- **<head>**: Contains meta-information and document settings.
- **<title>**: Sets the title shown in the browser tab.
- **<body>**: Contains the visible content of the page.
- **<div>**: Block-level container for content.
- ****: Inline container for text or other inline elements.
- ****: Unordered list
- ****: Ordered list
- ****: List item
- **<table>**: Defines a table

Common HTML Tags

- **<form>**: Defines an HTML form for user input
- **<input>**: Input control
- **<textarea>**: Multi-line text input control
- **<button>**: Button
- **<select>**: Dropdown list
- **<option>**: Option in a select list
- **<label>**: Defines a label for an **<input>** element
- **<tr>**: Table row
- **<td>**: Table data/cell
- **<th>**: Table header



Hello World Program in HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML Tutorial</title>
</head>
<body>
  <h1>Welcome To Software Fellowship Sponsored by TechAxis.
</h1>
  <p>Hello World!</p>
</body>
</html>
```

**Welcome To Software Fellowship Sp
TechAxis.**

Hello World!

EXERCISE 1

Create webpage for your favourite movie.

- Poster Image using `<img>` tag
- Movie Name using `<h1>` tag
- Production company using `<h2>` tag
- Movie Description using `<p>` tag
- Also, try attaching movie video from youtube. Use `<iframe>`



Lucky The Racer

by Goldmines

It's all about two brothers who do things differently. While one brother follows rules and regulations, the other tries to achieve it in his own way.



ASSIGNMENT 1

Create a login page which must contain email and password field with a submit button.

CSS

Overview

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About CSS

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CSS Syntax and Types

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CSS Selectors

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Common CSS Properties

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Display and Positioning

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CSS Units and
Responsive Design

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Assignment

About CSS

- CSS stands for Cascading Style Sheets.
- It is used to style and layout web pages.
- CSS allows you to control the color, font, spacing, and layout of HTML elements.
- It separates content (HTML) from design (CSS), enhancing maintainability.
- Enables responsive web design for different devices.

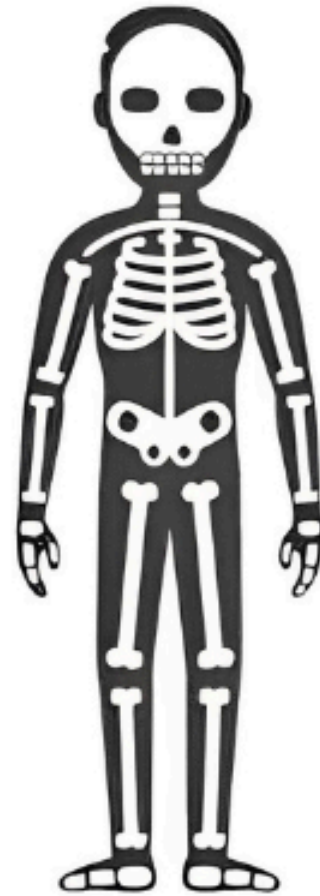
CSS



HTML



HTML the Skeleton



CSS



CSS the Skin



JS



Javascript the Brain



CSS Syntax

- CSS rules consist of selectors and declarations.
- Selector: Targets the HTML element(s) to style.
- Declaration: Contains property and value, specifying the style.
- Syntax: selector { property: value; }
- Multiple declarations can be grouped within the same selector.

```
<style>
body {
    background-color: lightgreen;
}

h1 {
    color: red;
    text-align: center;
}
</style>
```

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Types of CSS

- **Inline CSS:** Applied directly to HTML elements using the style attribute.
 - **Example:** `<p style="color: blue;">This is a blue paragraph.</p>`
- **Internal CSS:** Defined within a `<style>` tag inside the `<head>` section of an HTML document.
 - **Example:** `<head><style>p { color: green; } </style></head>`
- **External CSS:** Written in a separate .css file and linked to the HTML document using the `<link>` tag.
 - **Example:** `<link rel="stylesheet" href="styles.css">`

CSS Selectors

- **Element Selector: Selects HTML elements by their name.**
 - **Example:** `p { }` (styles all `<p>` elements)
- **Class Selector: Selects elements with a specific class, defined with a period ..**
 - **Example:** `.className { }` (styles elements with `class="className"`)
- **ID Selector: Selects a single element with a specific ID, defined with a hash #.**
 - **Example:** `#idName { }` (styles the element with `id="idName"`)
- **Universal Selector: Selects all elements, defined with an asterisk *.**
 - **Example:** `* { }` (styles all elements)
- **Attribute Selector: Selects elements with a specific attribute.**
 - **Example:** `[type="text"] { }` (styles elements with `type="text"`)

Common CSS Properties

- **Color:** Sets the text color. Example: `color: red;`
- **Background:** Sets the background color or image. Example: `background-color: blue;`
- **Font:** Sets font properties such as size, family, weight. Example: `font-size: 16px;`
- **Text Alignment:** Aligns text within an element. Example: `text-align: center;`
- **Margin:** Sets the outer spacing of an element. Example: `margin: 10px;`
- **Padding:** Sets the inner spacing of an element. Example: `padding: 20px;`
- **Display:** Specifies how an element should be displayed. Example: `display: block;`, `display: inline;`, `display: inline-block;`

Display Property

- `display: block;` Makes the element a block-level element, taking up full width.
- `display: inline;` Makes the element an inline-level element, only taking up as much width as necessary.
- `display: inline-block;` Combines features of block and inline, allowing for block-level styling with inline behavior.
- `display: none;` Hides the element completely from the page layout.



display: block

display:
block

display:
block

display:
block

display: inline

display: inline display:
inline display: inline

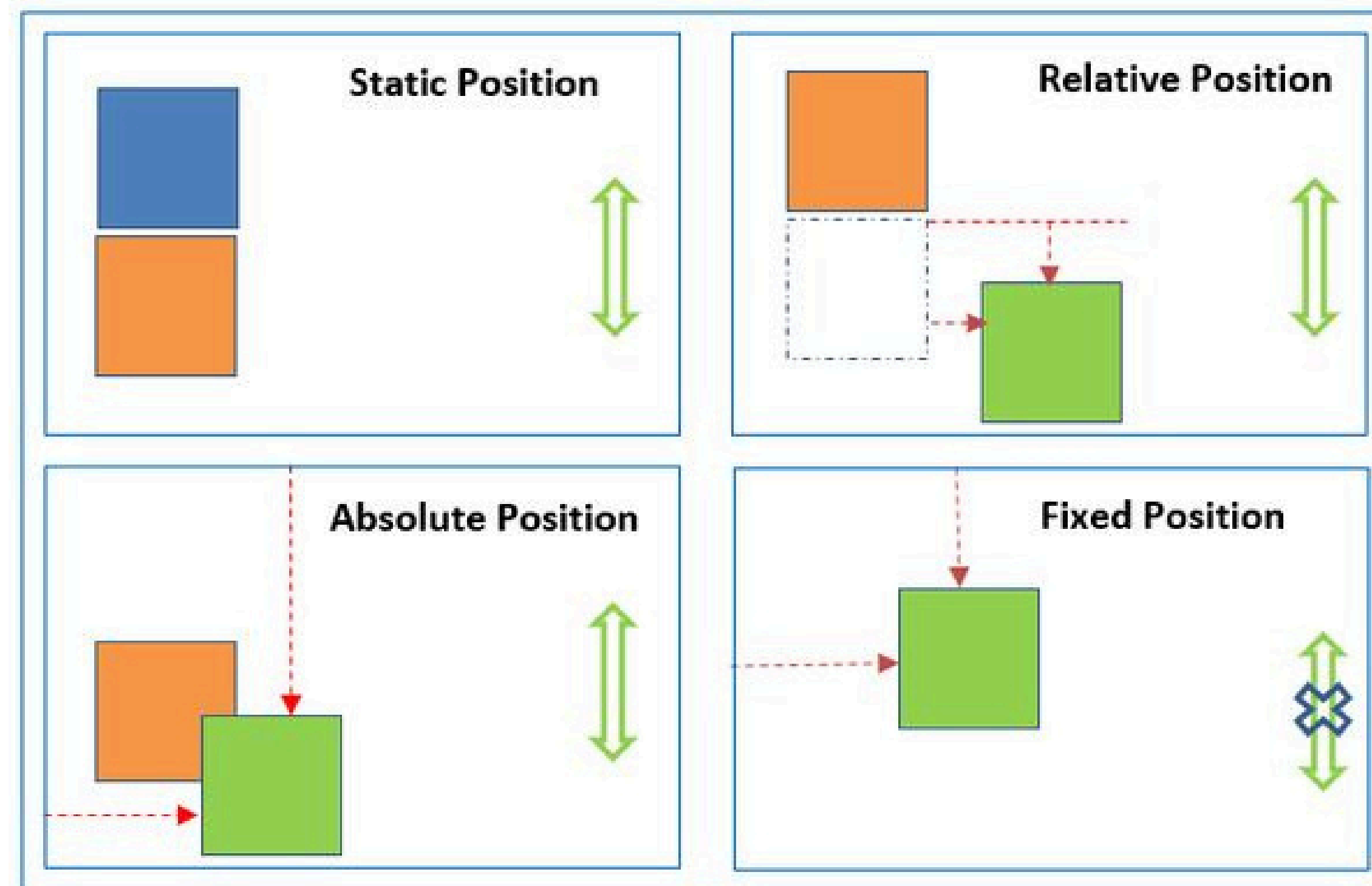
display: inline-block

display:
inline-
block

display:
inline-
block

display:
inline-
block

Positioning Property



Positioning Property

- `position: static;`: Default positioning behavior, elements flow according to document layout.
- `position: relative;`: Positions relative to its normal position, allowing for offset adjustments with top, right, bottom, left.
- `position: absolute;`: Positioned relative to its nearest positioned ancestor, ignoring document flow.
- `position: fixed;`: Positioned relative to the viewport, stays fixed even when scrolling.
- `position: sticky;`: Combination of relative and fixed positioning. Acts like relative positioning until it reaches a specified point, then acts like fixed.

Flexbox

- `display: flex;` **Turns an element into a flex container.**
- `flex-direction: row | row-reverse | column | column-reverse;` **Specifies the direction of the main axis.**
- `justify-content: flex-start | flex-end | center | space-between | space-around | space-evenly;` **Aligns items along the main axis.**
- `align-items: stretch | flex-start | flex-end | center | baseline;` **Aligns items along the cross axis.**
- `flex-wrap: nowrap | wrap | wrap-reverse;` **Controls whether flex items are forced into a single line or can wrap onto multiple lines.**

Flexbox Example

```
.flex-container {  
  display: flex;  
  flex-direction: row;  
  justify-content: space-between;  
  align-items: center;  
  flex-wrap: wrap; /* Allows ite  
}
```

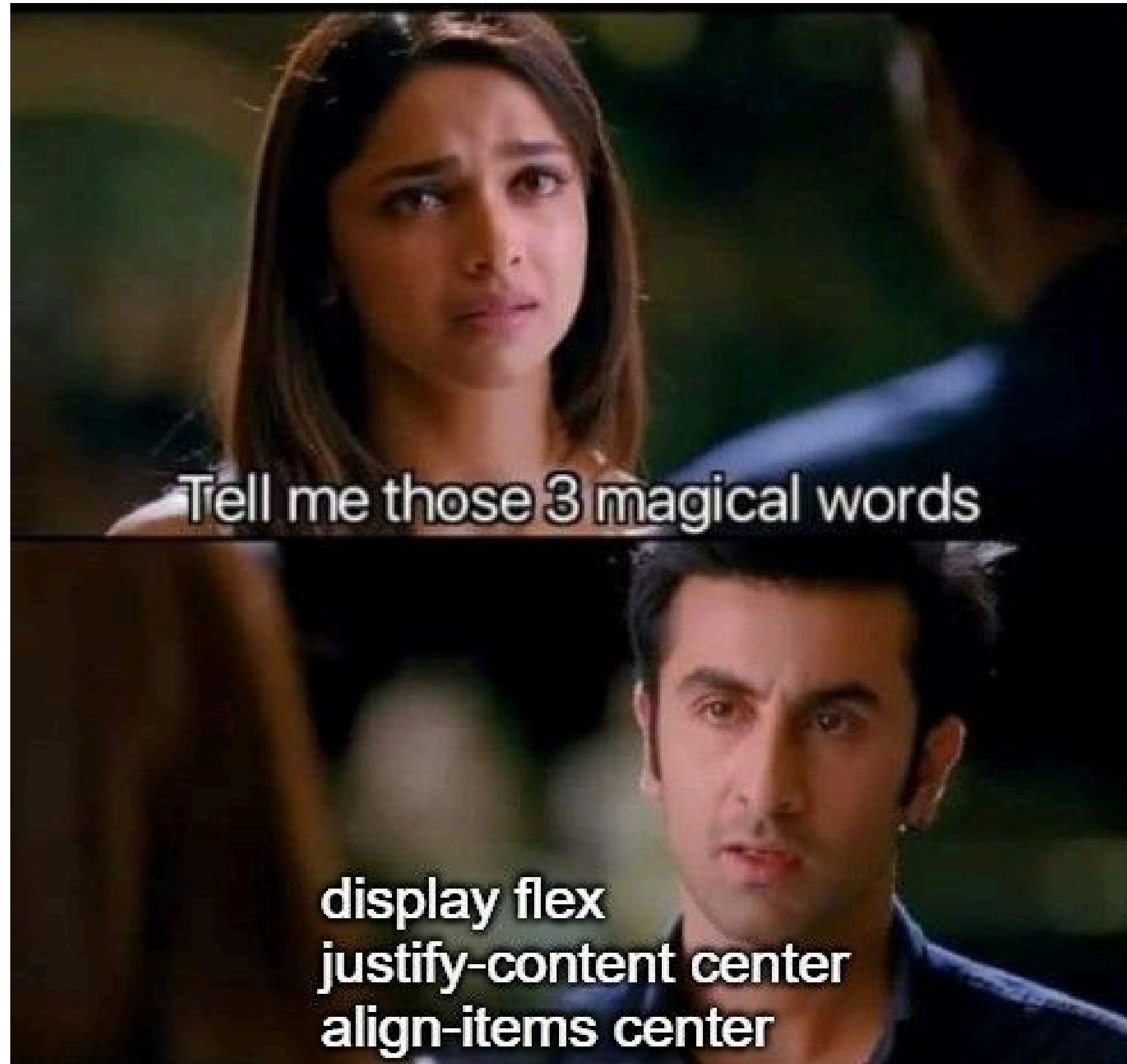
Software Fellowship

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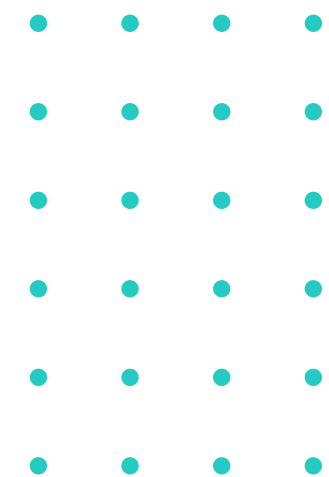
JOBAXLE



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Learn First To Lead The Rest



Optimization



Grid Layout

- `display: grid;` **Turns an element into a grid container.**
- `grid-template-rows: <track-size>;` **Defines the size of the rows in the grid.**
- `grid-template-columns: <track-size>;` **Defines the size of the columns in the grid.**
- `grid-template-areas: <grid-area-name>;` **Specifies named grid areas.**
- `grid-gap: <grid-row-gap> <grid-column-gap>;` **Sets the gap between rows and columns.**

Grid Layout Example

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CSS Units

- **Absolute Units:**

- px: Pixels, relative to the viewing device.
- pt: Points, 1/72 of an inch, used in print media.

- **Relative Units:**

- em: Relative to the font size of the element.
- rem: Relative to the font size of the root element (html).
- %: Percentage relative to the parent element.
- vw: Viewport width, 1% of the viewport width.
- vh: Viewport height, 1% of the viewport height.

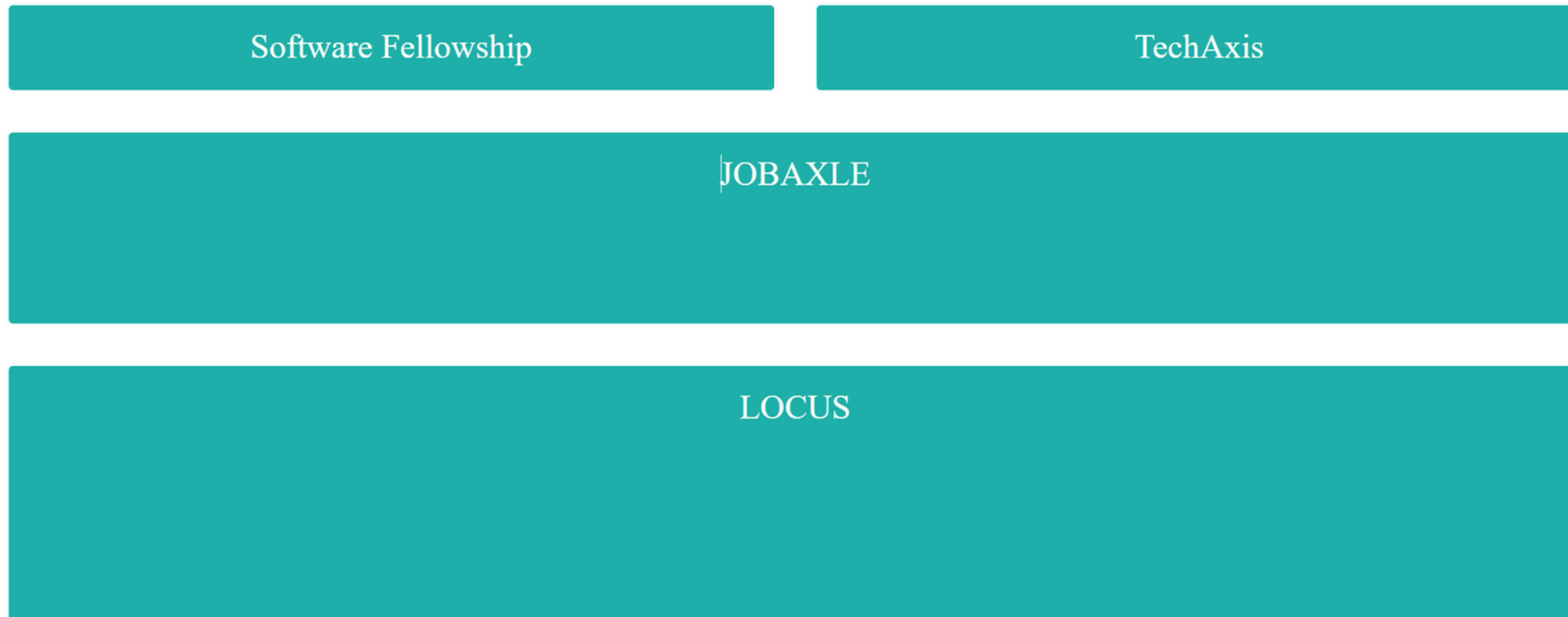
Responsive Design

- Techniques for making web pages look good on all devices.
- Media Queries: Apply different styles based on device characteristics.
- Flexible Grids: Use percentages or flexible units (em, rem) for layout.
- Flexible Images: Use `max-width: 100%;` to ensure images resize fluidly.

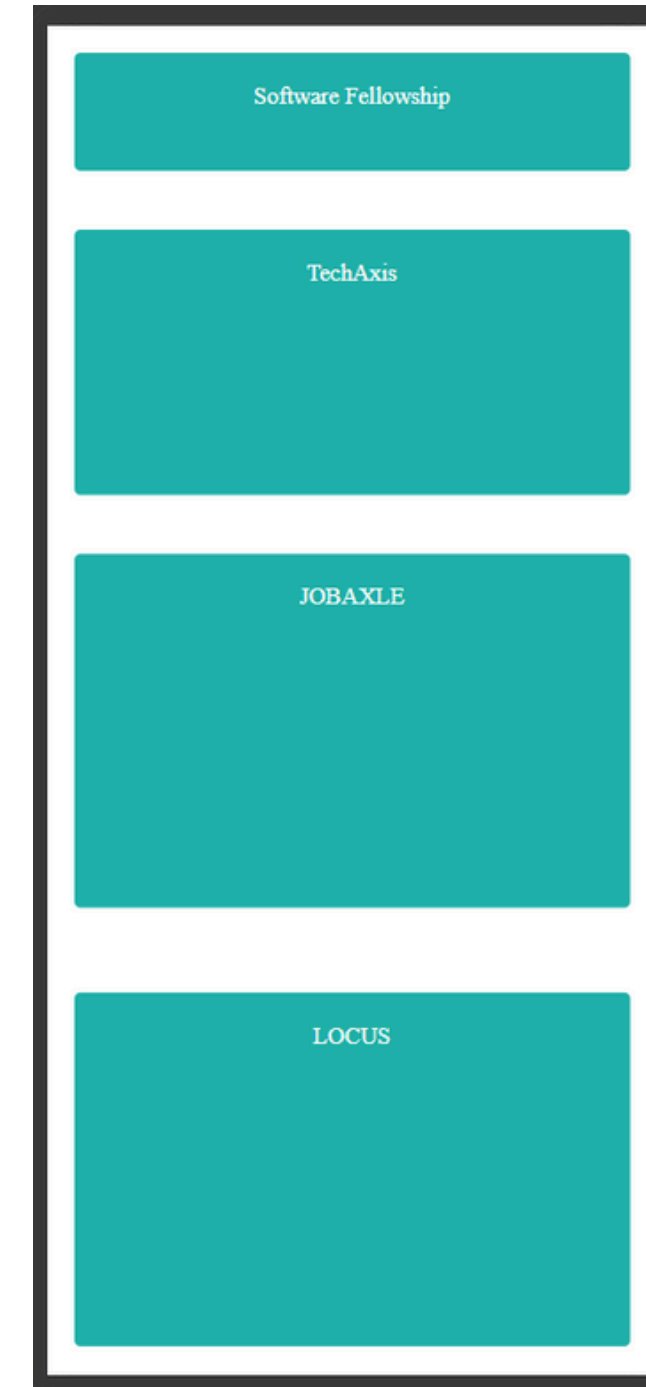
```
@media (max-width: 1024px) {  
  .grid-container {  
    grid-template-columns: 1fr;  
    grid-template-areas:  
      "area1"  
      "area2"  
      "area3"  
      "area4"; /* Adjusts the  
  }  
}
```

Responsive Design Example

Desktop view



Mobile View



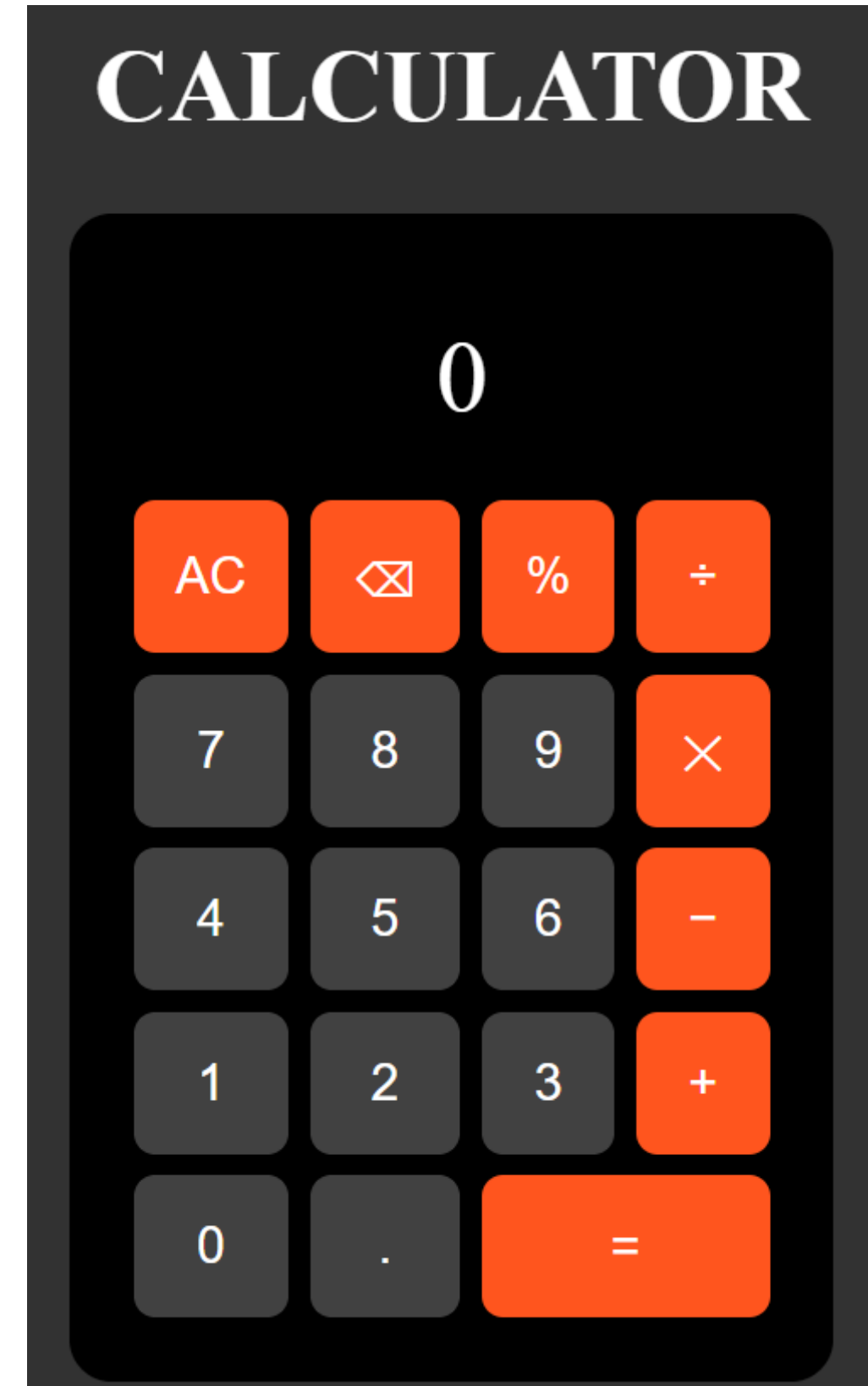
Resources for Learning CSS

- [MDN Web Docs](#)
- [W3Schools CSS Tutorial](#)
- [CSS-Tricks](#)
- [Learn CSS Layout](#)
- [Codecademy CSS Course](#)
- [freeCodeCamp CSS Course](#)

EXERCISE

Create a calculator design
using html and css.

*Note: You will implement js into it next day.



ASSIGNMENT 2

Use css to design login page (refer to **ASSIGNMENT 1**).

ASSIGNMENT 3 (Optional)

Try designing calendar for any month using
HTML and CSS

THANK YOU